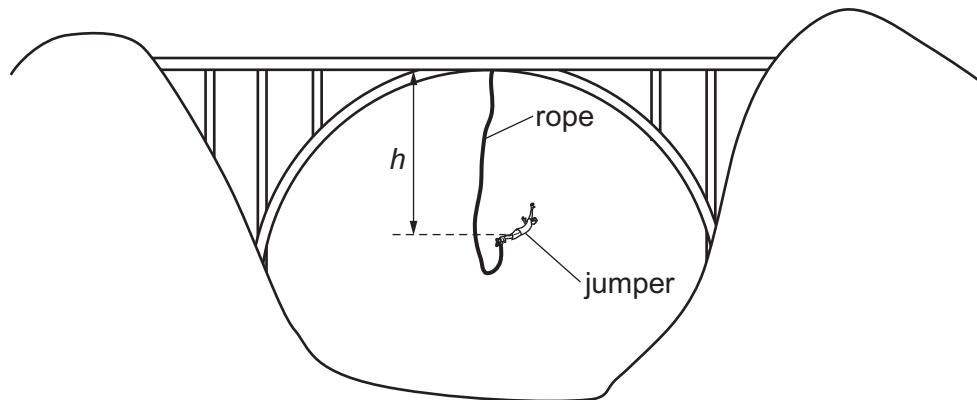
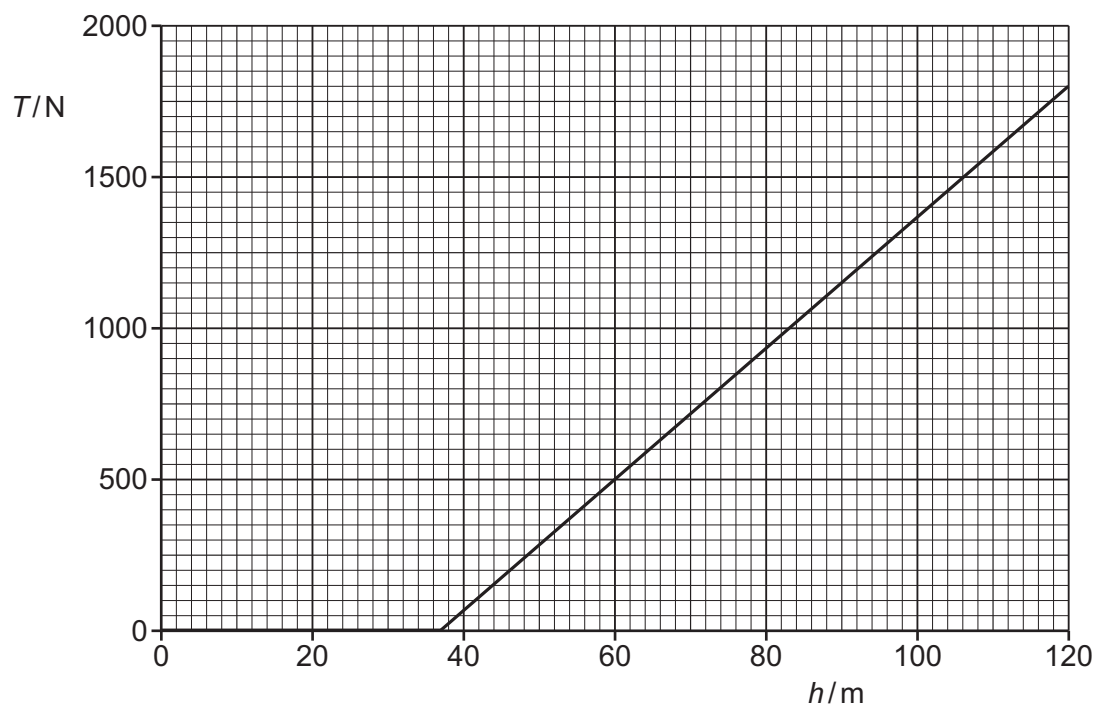


- 3 A bungee jumper of mass 64 kg secures one end of an elastic rope to a bridge. The other end is attached to the bungee jumper. The jumper falls from rest from the bridge and descends into the valley below, as shown in Fig. 3.1.



**Fig. 3.1** (not to scale)

Fig. 3.2 shows the variation of the tension  $T$  in the rope with the vertical distance  $h$  of the jumper below the level of the bridge.



**Fig. 3.2**

- (a) The rope obeys Hooke's law.

State Hooke's law.

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 ..... [1]

